

Rocksdb_dataset-7

Commit 121:

3740fccc4b4e85c0d012c7145a64168a2c23fca0

Non-C file modifications, therefore no performance regression testing is required.

Commit 122:

3af905aa685fc50586dbd6af54142daa11d24bdd

The 1st symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2034306.7818958333
4  The new version is executed by default
5  2030870.0315918368
6  The new version
7  2037059.2346956523
8  Comparison of New and old versions (along symbolic execution path)
9  -0.0013511893777748568
10 Comparison of New and old versions (along the default execution path)
11 -0.0030383029606599237
```

No performance regression

Commit 123:

d1f383b8ebd0e2ea746cbe538b29ca1a7576e9cc

Unable to provide test cases

Commit 124:

ebaeb036486053aeb86a721f5a3973981e2f05e4

Non-C file modifications, therefore no performance regression testing is required.

Commit 125:

1d6c33d2a59f6f9dd23e7db61618a0f0aa3c1600

Non-C file modifications, therefore no performance regression testing is required.

Commit 126:

7e272d20329449a7eb8d158d283e1635d029909a

Non-C file modifications, therefore no performance regression testing is required.

Commit 127:

0c7e5bd2f0d73d96f620f912a4569ca5375d1786

Non-C file modifications, therefore no performance regression testing is required.

Commit 128:

5f9b7ccce32ecc05214461c160b0f13644582be8

Non-C file modifications, therefore no performance regression testing is required.

Commit 129:

da8eba8b4927ae92be24b4dab3417187ca979d66

Unable to provide test cases

Commit 130:

15873b1fdd3a4aa7a21825616623e4fb482443e7

The 1st symbolic execution

Testcase:for(int i = 0; i< 1000; i++){db->SetOptions({{"write_buffer_size", "131072"}});}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  675862.1116595744
4  The new version is executed by default
5  673201.5216170213
6  The new version
7  686814.2006938775
8  Comparison of New and old versions (along symbolic execution path)
9  -0.015946218093974132
10 Comparison of New and old versions (along the default execution path)
11 -0.019820031477368348
```

No performance regression

The 2nd symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  271159.08925
4  The new version is executed by default
5  267279.76613043476
6  The new version
7  265536.3897755102
8  Comparison of New and old versions (along symbolic execution path)
9  0.021174873542731315
10 Comparison of New and old versions (along the default execution path)
11 0.006565489409562521
```

No performance regression

The 3rd symbolic execution

```
Testcase:for(int i = 0; i < 1000; i++){db->SetOptions({"write_buffer_size", "131072"});}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  670781.1797291668
4  The new version is executed by default
5  665658.8021702127
6  The new version
7  676538.526875
8  Comparison of New and old versions (along symbolic execution path)
9  -0.008510006329465022
10 Comparison of New and old versions (along the default execution path)
11 -0.016081456225474404
```

No performance regression

The 4th symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2032949.8795510202
4  The new version is executed by default
5  2039831.2642040818
6  The new version
7  2035545.5936041668
8  Comparison of New and old versions (along symbolic execution path)
9  -0.0012751932756026024
10 Comparison of New and old versions (along the default execution path)
11 0.002105416166250944
```

No performance regression

The 5th symbolic execution

```
Testcase:for(int i = 0; i< 1000000; i++){db->SetDBOptions({"max_background_flushes", "2"});}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1813771.8330612245
4  The new version is executed by default
5  1844431.995086957
6  The new version
7  1800331.6342040817
8  Comparison of New and old versions (along symbolic execution path)
9  0.007465401708105114
10 Comparison of New and old versions (along the default execution path)
11 0.024495687375049566
```

There may be performance regression

The 6th symbolic execution

```
Testcase:for (int i = 0; i < 10000; i++){key = "key_exit" + std::to_string(i + 30001);value =
"new_value" + key;db->Put(write_options_sync, key, value);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  349157.2068510638
4  The new version is executed by default
5  354809.23741666664
6  The new version
7  346384.0299166667
8  Comparison of New and old versions (along symbolic execution path)
9  0.00800607618966819
10 Comparison of New and old versions (along the default execution path)
11 0.02432331392999528
```

There may be performance regression

The 7th symbolic execution

Testcase:for (int i = 0; i < 10000; i++){key = "key_exit" + std::to_string(i + 30001);value = "new_value" + key;db->Put(write_options_sync, key, value);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  345554.3734791667
4  The new version is executed by default
5  354256.15654166666
6  The new version
7  344392.93848936167
8  Comparison of New and old versions (along symbolic execution path)
9  0.003372412323259407
10 Comparison of New and old versions (along the default execution path)
11 0.02863943173622785
```

There may be performance regression

Commit 131:

ec8f1452f5214f554f4d492aa33be83555202494

Non-C file modifications, therefore no performance regression testing is required.

Commit 132:

f6bff87b92577e5f0f2d8289a736103682d27070

Non-C file modifications, therefore no performance regression testing is required.

Commit 133:

14c949df8ba25bb8518b4283f9dd7e002bdf8e4a

Unable to provide test cases

Commit 134:

8e6d4311533d371f70993dd5b7441082ae2f3cd9

Unable to provide test cases

Commit 135:

68b2d941be2d273067aeb7e14ccd090d1512c426

The 1st symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2037320.1266086954
4  The new version is executed by default
5  2042941.7703469389
6  The new version
7  2035729.5814285716
8  Comparison of New and old versions (along symbolic execution path)
9  0.0007813145688081033
10 Comparison of New and old versions (along the default execution path)
11 0.003542803024607086
```

No performance regression

Commit 136:

d033c6a84958711b33f83ca3e9f40cf1ca57775a

Unable to provide test cases

Commit 137:

5d1c0a8832ce5aeb3b7c76044b079d079143dd73

Non-C file modifications, therefore no performance regression testing is required.

Commit 138:

b9c7481fc2424ac31a97929edc6dbad360be2646

The 1st symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  269627.23038775515
4  The new version is executed by default
5  272455.71539583337
6  The new version
7  271408.34308695653
8  Comparison of New and old versions (along symbolic execution path)
9  -0.006562483227093494
10 Comparison of New and old versions (along the default execution path)
11 0.0038590276811839327
```

No performance regression

Commit 139:

60c266658dc6597d4afce59297b9ebcd2998cf3a

Non-C file modifications, therefore no performance regression testing is required.

Commit 140:

22ca6e5e68381915ad317db9e0afc06a52742fcd

Unable to provide test cases